

SENTINEL AI — DIGITAL FORENSICS & SOC PLATFORM

Comprehensive Technical Guide: Role Functions, Exact Inputs, Expected Outputs & Operational Architecture

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Target Audience: SOC Analysts, Forensic Investigators, Admins	AI Engine: Google Gemini 1.5 Flash

1. System Architecture & Operational Overview

SENTINEL AI is an enterprise-grade Digital Forensics and Incident Response (DFIR) platform combined with a Security Operations Center (SOC) Workspace. The system parses multi-vector evidence streams in real time—including RFC-822 emails, network PCAP binaries, PE/ELF executables, domain WHOIS records, and SIEM event streams—correlated live with the **Google Gemini AI Engine**.

The architecture enforces strict **Role-Based Access Control (RBAC)** so that each security professional operates in a streamlined, uncluttered environment tailored to their exact workflow. All computations run on live dynamic execution engines without pre-packaged static fallbacks.

2. Role-Based Access Control (RBAC) Matrix

The platform segregates duties into three distinct operational roles. Below is the functional permission and visibility matrix:

Feature / Module	SOC Analyst	Forensic Investigator	System Admin
Analyst Workspace (/analyst) (Triage, YARA, SIEM Parser, SOAR)	FULL ACCESS	Hidden & Restricted	FULL ACCESS
Case Management (/cases) (Case Creation, Evidence Attachment)	Hidden & Restricted	FULL ACCESS	FULL ACCESS
PDF Investigation Reports (/reports)	Hidden & Restricted	FULL ACCESS	FULL ACCESS
Standalone Forensics Tools (Email, URL, PCAP, Malware, Threat Intel)	FULL ACCESS	FULL ACCESS	FULL ACCESS

Gemini AI Chat Assistant	FULL ACCESS	FULL ACCESS	FULL ACCESS
Admin Control Panel (/admin)	Hidden & Restricted	Hidden & Restricted	FULL ACCESS

3. Complete Module Input & Output Specifications

This section details the exact input required for every tool in each role, alongside the precise structured output generated by the engine.

3.1 Email Forensics Module (/email-forensics)

Email RFC-822 Header & Attachment Analyzer	ROLE: Analyst, Investigator, Admin
REQUIRED INPUT FORMAT:	Upload raw <code>`EML`</code> or <code>`MSG`</code> email file or paste raw RFC-822 MIME headers into text input area.
INPUT EXAMPLE:	From: CEO <ceo@company.com> To: Finance <finance@company.com> Subject: Urgent Wire Transfer Received: from mail.attacker.com (185.220.101.5)
GENERATED OUTPUT SUMMARY:	Parses email headers, verifies authentication signatures, computes attachment checksums, and assesses phishing risk.
STRUCTURED OUTPUT FIELDS:	• Hop Trace: Array of IP hops, timestamps, and server hostnames. • Auth Checks: SPF (Pass/Fail), DKIM signature validity, DMARC alignment. • Attachment Hashes: MD5 & SHA256 cryptographic hashes of attachments. • Risk Score: 0–100 Threat Rating (CRITICAL / HIGH / MEDIUM / LOW).

3.2 URL Threat Analysis Module (/url-forensics)

Live URL & Domain Infrastructure Analyzer	ROLE: Analyst, Investigator, Admin
REQUIRED INPUT FORMAT:	Type or paste any target URL, domain name, or IP address into the search input box.
INPUT EXAMPLE:	<code>https://login.paypal-security-update.account-verify.xyz/login.php</code>
GENERATED OUTPUT SUMMARY:	Executes live DNS resolution, WHOIS registrar queries, SSL certificate verification, and brand spoofing detection.
STRUCTURED OUTPUT FIELDS:	• Resolved IP & Location: Server IP address, country, ISP, and hosting provider. • Domain Age: Days since registration (flags domains < 30 days old). • Shannon Entropy: Mathematical randomness score of URL path. • Brand Spoofing Flag: Identifies typosquatting target (e.g. PayPal, Google, Microsoft).

3.3 Network PCAP Forensics Module (/network-forensics)

Binary PCAP Network Stream Dissector	ROLE: Analyst, Investigator, Admin
REQUIRED INPUT FORMAT:	Upload binary <code>.pcap</code> or <code>.pcapng</code> packet capture file (max 50MB per upload).
INPUT EXAMPLE:	<code>sample_capture.pcap</code> (binary Ethernet/IP frames containing TCP/UDP/DNS flows)
GENERATED OUTPUT SUMMARY:	Dissects packet layers, reconstructs TCP conversations, flags beaconing IPs, and extracts DNS domain queries.
STRUCTURED OUTPUT FIELDS:	• Bandwidth & Packet Metrics: Total packets, total bytes, duration. • Protocol Breakdown: HTTP, HTTPS, DNS, SSH, FTP packet ratios. • Top IP Talkers: Source/Destination IPs ordered by packet volume. • Extracted DNS Queries: List of resolved domain names from DNS frames.

3.4 Malware Forensics Module (/malware-forensics)

PE / ELF Static Binary Analysis Engine	ROLE: Analyst, Investigator, Admin
REQUIRED INPUT FORMAT:	Upload executable binary file (.exe, .dll, .bin, .elf, .apk, .pdf).
INPUT EXAMPLE:	update_svc.exe (binary Windows Portable Executable file)
GENERATED OUTPUT SUMMARY:	Computes entropy across file sections, extracts Win32 API import flags, and identifies malware packing.
STRUCTURED OUTPUT FIELDS:	• File Hashes: MD5 & SHA-256 binary checksums. • Entropy Score: 0.00–8.00 rating (Entropy > 7.2 flags packing/encryption). • Suspicious API Imports: Flags APIs like VirtualAllocEx, WriteProcessMemory, CreateRemoteThread. • Malware Family Hint: Heuristic classification (e.g., Ransomware, Spyware, Trojan).

3.5 Threat Intelligence Module (/threat-intelligence)

Multi-Feed Threat Intel IOC Lookup	ROLE: Analyst, Investigator, Admin
REQUIRED INPUT FORMAT:	Enter any IP address, domain name, file hash (MD5/SHA256), or CVE ID.
INPUT EXAMPLE:	185.220.101.5 OR c5b3a...4f2e OR attacker-c2.xyz
GENERATED OUTPUT SUMMARY:	Queries multi-feed threat intelligence databases for reputation scores and campaign tagging.
STRUCTURED OUTPUT FIELDS:	• Reputation Score: Malicious detection ratio across threat feeds. • Known Campaigns: Associated APT groups or malware operations. • Geolocation & ASN: Autonomous System Number and host organization.

3.6 Analyst SOC Workspace (/analyst)

Dedicated workspace for Security Operations Center Analysts containing 5 interactive sub-tools:

Sub-Tool	Input Required	Output Generated
1. Alert Triage Queue	Click action buttons: FALSE POSITIVE, CONTAIN IP, or ESCALATE on incoming alerts.	Alert status changes immediately. ESCALATE automatically creates an official investigation case.
2. Threat Hunting Workbench	• Select YARA or SIGMA mode. • Paste Rule Code in textarea. • Paste Test Log/Payload sample.	Signature validation report: Match Status (MATCH_FOUND / NO_MATCH), match count, and matched line snippets.
3. SIEM Log Parser	Paste raw Syslog, EVTX, or JSON log text stream into log input box.	Structured anomaly summary: Lines parsed, Failed login count, Shell execution events, and Top IP talkers.

4. Attack Surface Scanner	Type target IP address or Domain name into search box.	Security Health Grade (A–F), overall risk score, open ports table, and matching CVE vulnerabilities.
5. SOAR Playbook Engine	Select action type (Block IP, Sinkhole Domain, Isolate Endpoint) and enter Target Value.	Executable 1-click containment payloads: Linux iptables / BIND scripts and Windows PowerShell firewalls.

3.7 Case Management & Incident Reports (/cases & /reports)

Incident Case Creator & PDF Report Generator	ROLE: Investigator, Admin
REQUIRED INPUT FORMAT:	- Case Title, Description, Category, Threat Level (CRITICAL / HIGH / MEDIUM / LOW). - Upload evidence artifacts to case.
INPUT EXAMPLE:	Title: 'Corporate Mail Server Ransomware Intrusion' Threat Level: CRITICAL Category: Malware
GENERATED OUTPUT SUMMARY:	Unified case file, automated attack timeline, evidence correlation matrix, and downloadable executive PDF report.
STRUCTURED OUTPUT FIELDS:	Case Number: Unique ID (e.g. CASE-2026-9041). Unified Timeline: Chronological sequence of all uploaded artifacts. Executive Report PDF: Printable PDF containing executive summary, IOC table, and remediation steps.

4. Step-by-Step Operator Workflow Guide

Step 1: Logging In & Role Verification

Navigate to /login and log in with your account credentials. Check the bottom of the left sidebar to verify your assigned role (**ANALYST**, **INVESTIGATOR**, or **ADMIN**).

Step 2: Analyst Triage Workflow (For SOC Analysts)

- Go to **Analyst Workspace** (/analyst).
- In the **Alert Triage Queue**, review incoming alerts. Click FALSE POSITIVE for noise, or CONTAIN IP to generate block scripts.
- Click ESCALATE on high-severity alerts to create an official case.
- Use the **SIEM Log Parser** or **Threat Hunting Workbench** to validate YARA/Sigma detection rules against new payloads.

Step 3: Investigation Workflow (For Forensic Investigators)

- Go to **Cases & Incidents** (/cases).
- Click **NEW INVESTIGATION CASE** and fill in incident details.
- Open the case details and upload evidence into Email, URL, PCAP, or Malware tabs.
- Click **GENERATE PDF REPORT** to export an official forensic report.

Step 4: AI Correlation & System Administration (For Admins)

1. Configure your Google Gemini API key in **Admin Control Panel** (/admin).
2. Monitor platform audit trail logs and switch user roles as needed.